Bode Part 1

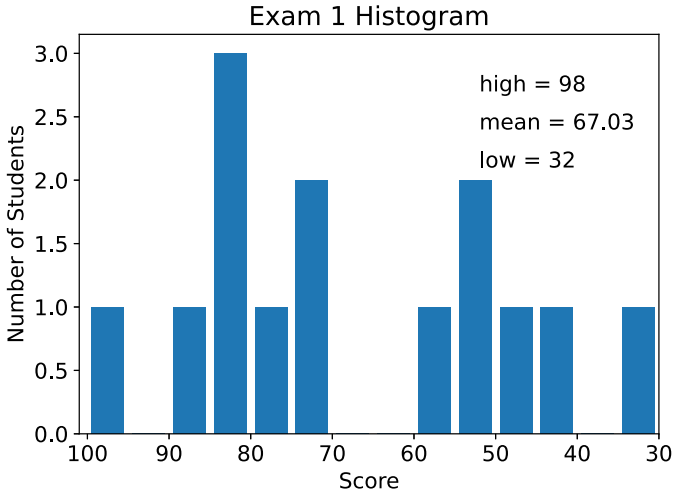
EGR 345

GVSU

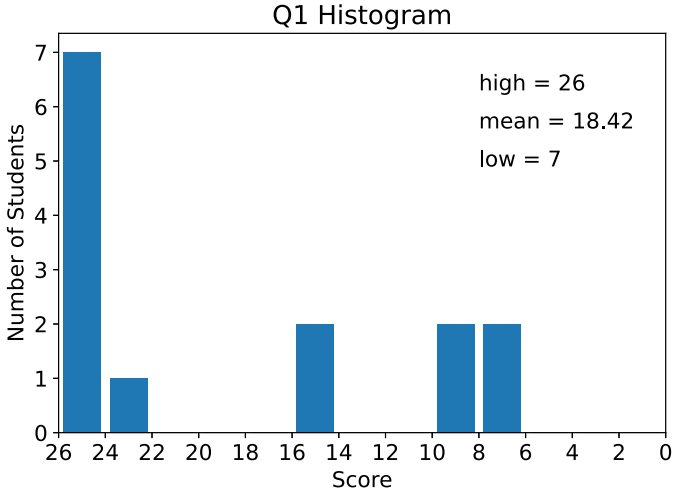
Outline

- exam 1
  - histograms
  - solution
  - my grading philosophy
- pendulum labs big picture
  - what were you supposed to learn?
- Bode Part 1

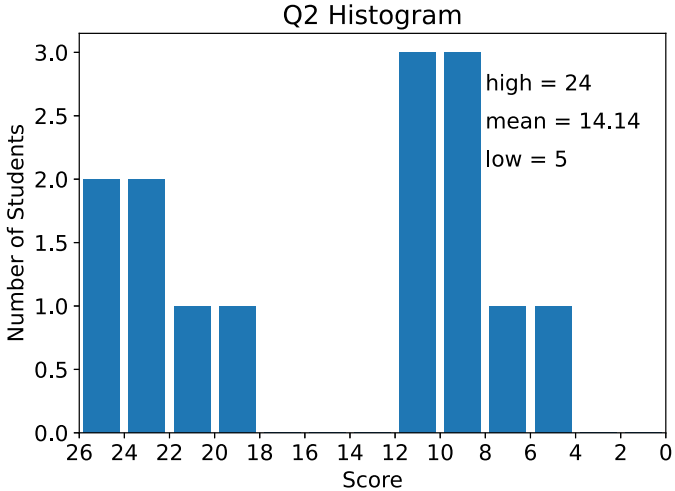
Overall Histogram



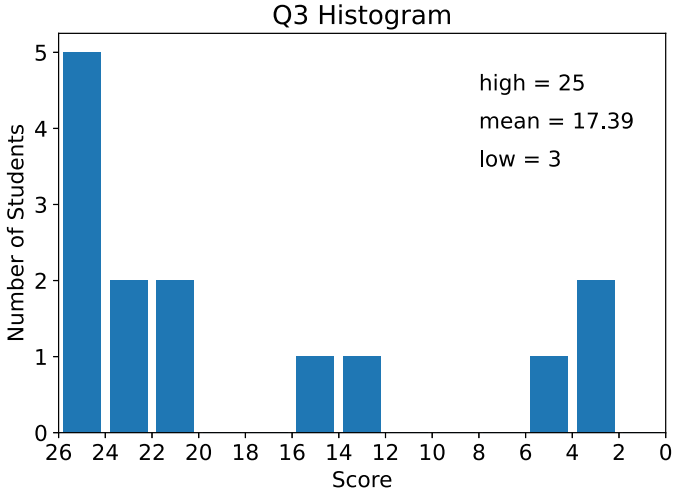
Question 1 Histogram



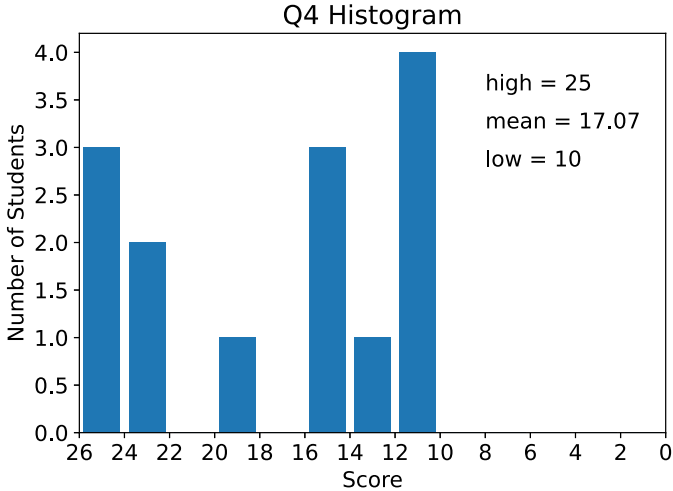
Question 2 Histogram



Question 3 Histogram



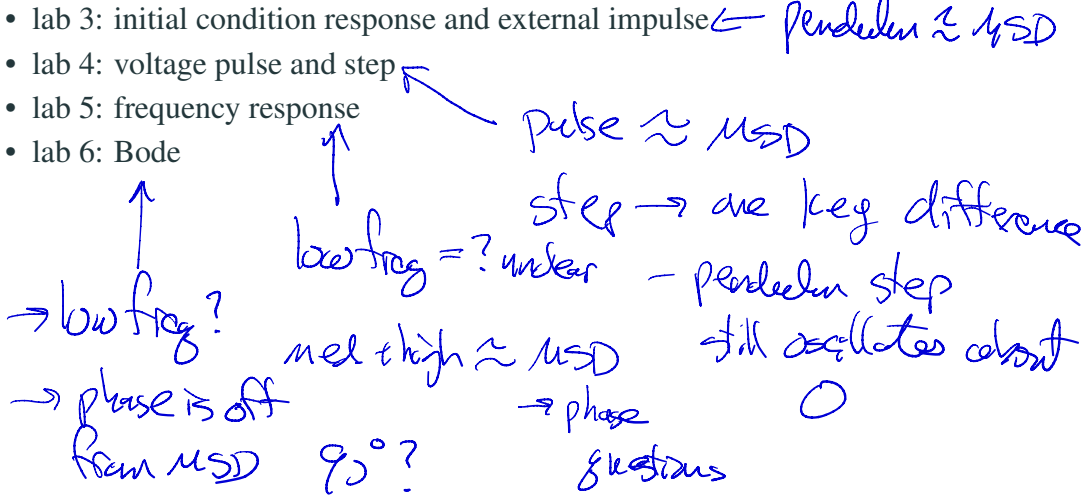
Question 4 Histogram



Pendulum Labs Big Picture

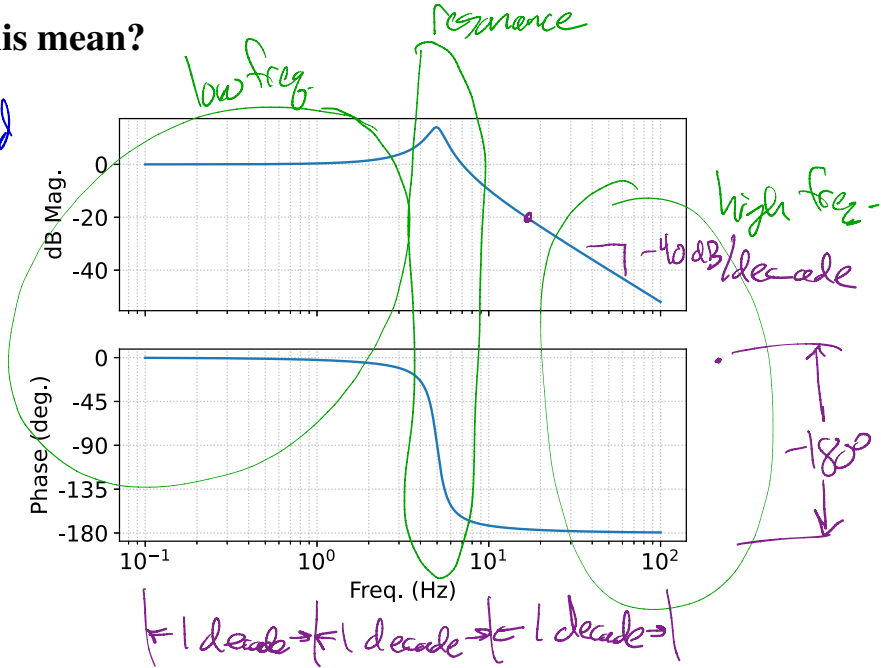
What were you supposed to learn from the pendulum labs?

- lab 3: initial condition response and external impulse
- lab 4: voltage pulse and step
- lab 5: frequency response
- lab 6: Bode



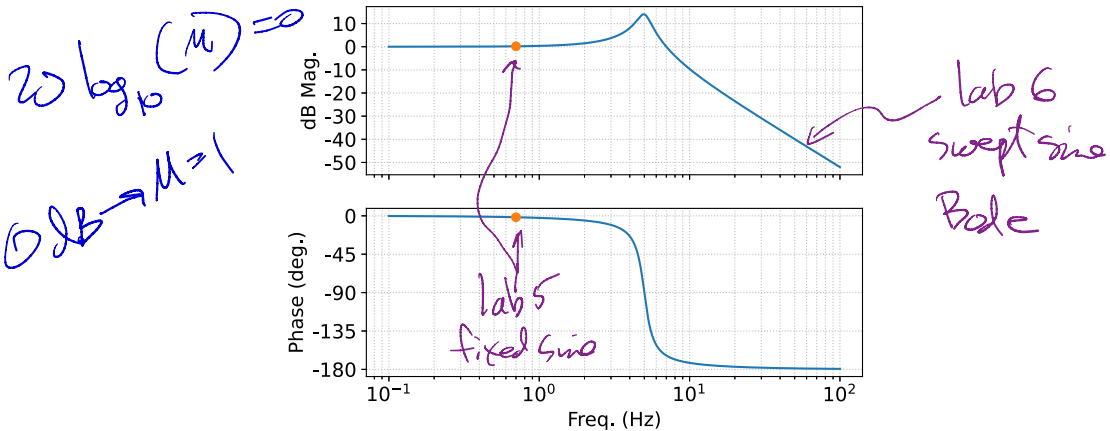
Bode Part 1

What does this mean?

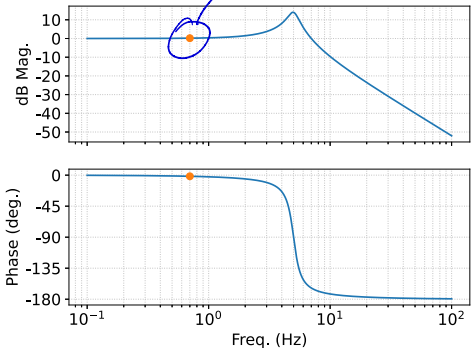


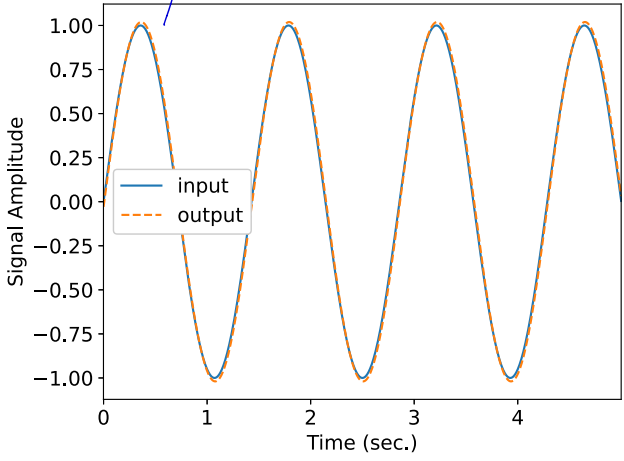
Fixed Sine vs. Bode 1

- what would you expect the steady-state input/output relationship to look like?



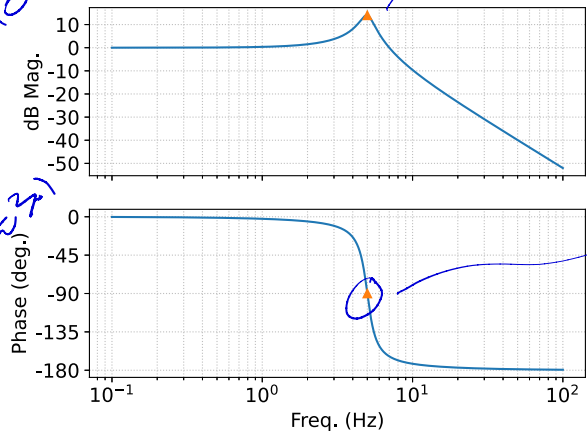
Fixed Sine vs. Bode 1 (answer)



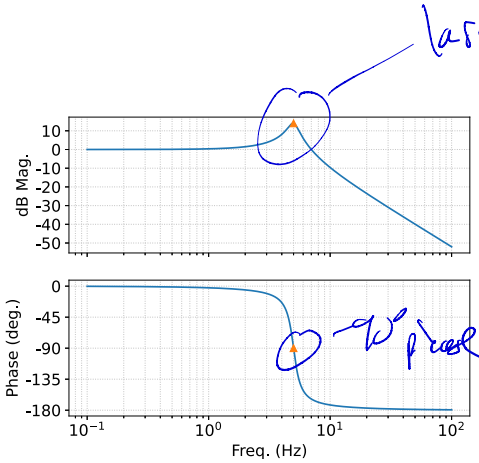


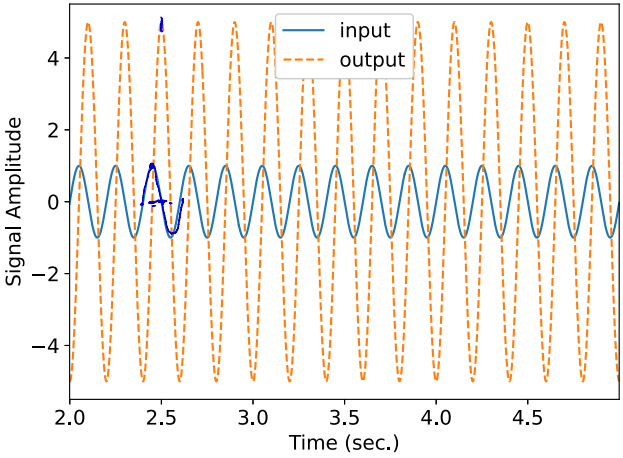
Fixed Sine vs. Bode 2

- what would you expect the steady-state input/output relationship to look like?



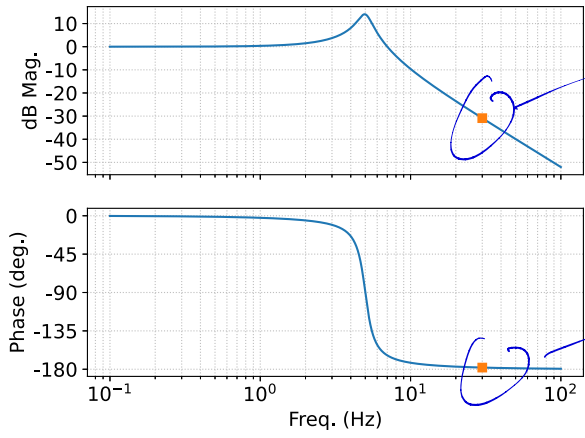
Fixed Sine vs. Bode 2 (answer)



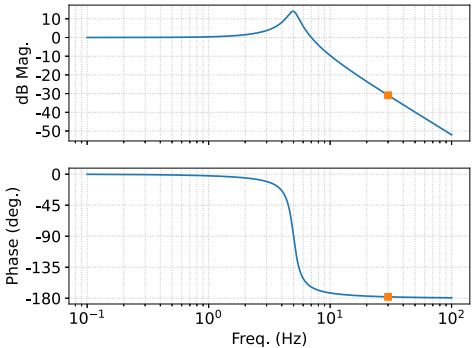


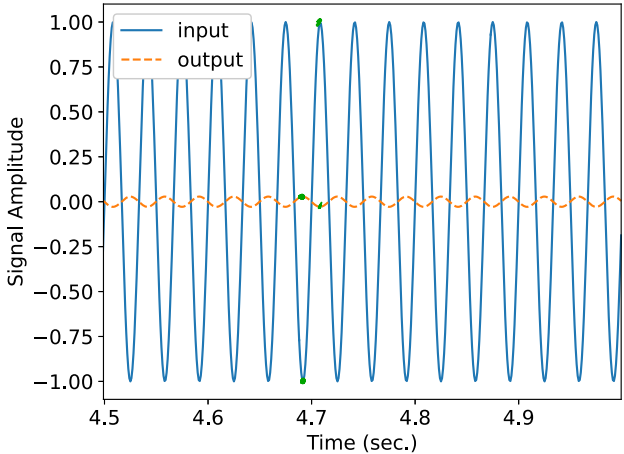
Fixed Sine vs. Bode 3

- what would you expect the steady-state input/output relationship to look like?



Fixed Sine vs. Bode 30 (answer)





Analyzing Bode Frequency Ranges

$$G(s) = \frac{\omega^2}{s^2 + 2\zeta\omega_n s + \omega^2}$$

$$G(j\omega) =$$

- low frequency
- near resonance
- high frequency